



Safety fundamentals of GenAI

Logistics and Course Overview

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Teaching assistants: Naquee Rizwan, Sai Keerthana, Paramananda Bhaskar

Course website: <https://sites.google.com/view/safe-gen-ai/>



What do you think of, when hearing "Safe AI"?

- phone no.s
credit
card
details
- ① Privacy [encrypted data should remain encrypted.]
 - ② Does not allow generation of harmful content.
 - ③ Hallucination

What does AI Safety Mean Anyway?

Helen Toner defines AI Safety as follows: *“AI safety focuses on technical solutions to ensure that AI systems operate safely and reliably.”*

AI safety

Tim G. J. Rudner and Helen Toner, "Key Concepts in AI Safety: An Overview" (Center for Security and Emerging Technology, March 2021). <https://doi.org/10.51593/20190040>.

https://medium.com/@tenyks_blogger/what-is-ai-safety-neurips-2023-series-31e2c65256ed

What does AI Safety Mean Anyway?

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AI safety

- Is an interdisciplinary field concerned with preventing accidents, misuse, or other harmful consequences that could result from AI systems
- Encompasses machine ethics and AI alignment (which aim to make AI systems moral and beneficial)
- Monitoring AI systems for risks and making them highly reliable.
- Involves developing norms and policies that promote safety.

CS & Social sciences
& safety for liability
stuff.

GenAI
↳ not only
problem solving
units
ethical & moral
in making
decisions

What does AI Safety Mean Anyway?

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AI safety

- Is an **interdisciplinary field** concerned with **preventing** accidents, misuse, or other **harmful consequences** that could result from AI systems
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What is not AI Safety

AI safety addresses technical challenges AND ethical, societal, and long-term considerations.

Three Common Misconceptions of AI Safety

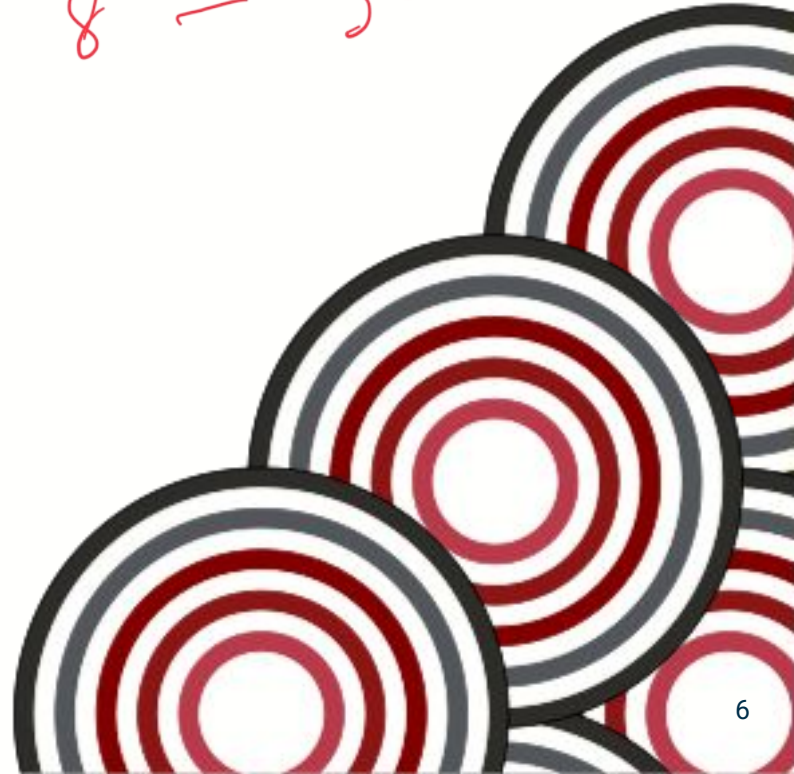
- **Not Just Security Measures:** While security is a part of it, AI safety encompasses a broader scope, including ethical considerations, responsible development practices, and the prevention of unintended consequences.
- **Not Limited to Technical Aspects:** It's not just about ensuring algorithms work correctly but also about understanding and addressing the broader impacts on individuals and communities.
- **Not Solely About Avoiding Malicious Use:** AI safety involves designing systems to be robust, interpretable, and aligned with human values. It's not just about avoiding intentional harm but also about minimizing unintentional negative impacts.

Tim G. J. Rudner and Helen Toner, "Key Concepts in AI Safety: An Overview" (Center for Security and Emerging Technology, March 2021). <https://doi.org/10.51593/20190040>.
https://medium.com/@tenyks_blogger/what-is-ai-safety-neurips-2023-series-31e2c65256ed

What Do You Feel Positive About AI?

Automation
Companionship

8 - 30



What Do You Feel Negative About AI?

impersonating someone.
Are your jobs at stake?
Frauds → increased in presence of AI
deep fakes

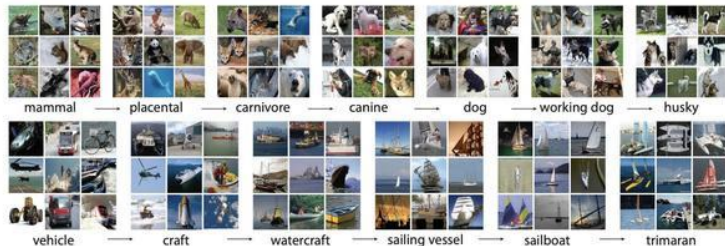
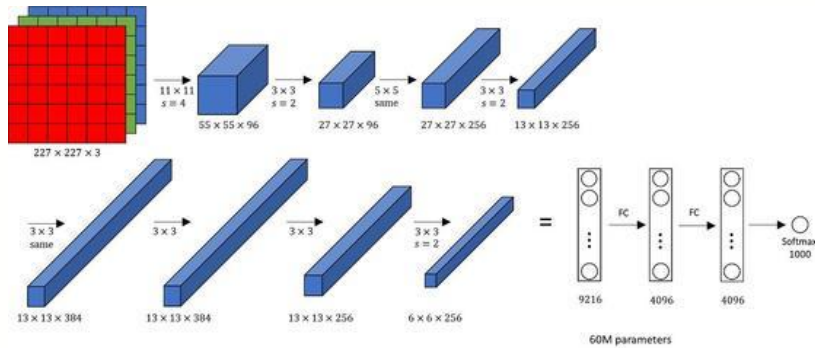


How Positive/Negative Do You Feel About AI?



Success of Deep Learning

Image classification: Deep Learning breakthrough
with AlexNet in 2012



Resnet.

↓

image classification

pre req of
either
NLP (CS)
or
DL.

groups -
semantic
groups

Success of Deep Learning

Image classification: Deep Learning breakthrough
with AlexNet in 2012

Reinforcement learning:



Human vs ~~AP~~
~ Go game

protein structures
AlphaFold was proposed.
AlphaGo

Success of Deep Learning

Image classification: Deep Learning breakthrough with AlexNet
in 2012

Reinforcement learning:

Alpha GO beat Lee Sedol in a five-game match in 2016 ✓

Alpha Fold revolutionizes protein structure prediction in 2018 ✓



Success of Deep Learning (Selection)

Image classification: Deep Learning breakthrough with AlexNet

in 2012

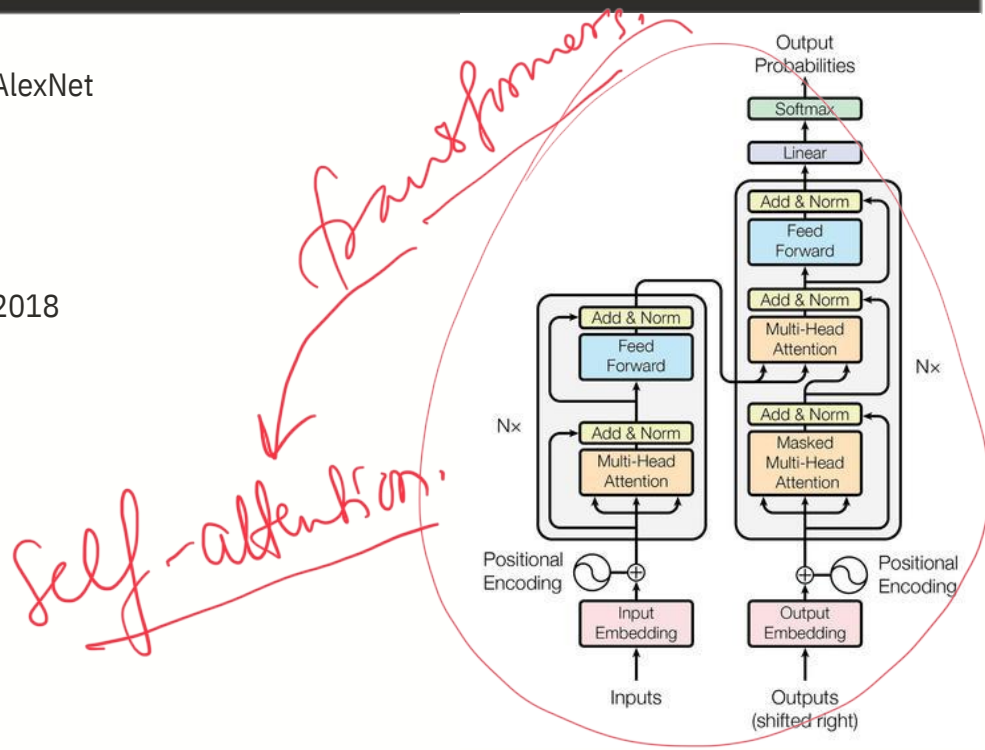
Reinforcement learning:

Alpha GO beat Lee Sedol in a five-game match in 2016

Alpha Fold revolutionizes protein structure prediction in 2018

Natural Language Processing:

Transformer architecture paper in 2017



NIPS papers
<https://papers.nips.cc/paper/7181-attention...>

Attention is All you Need

by A Vaswani - Cited by 115085 — We propose a new simple network architecture, the **Transformer**, based solely on attention mechanisms, dispensing with recurrence and...

<https://arxiv.org/abs/1706.03762>

Success of Deep Learning (Selection)



Image classification: Deep Learning breakthrough with AlexNet

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Alpha GO beat Lee Sedol in a five-game match in 2016

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Natural Language Processing:

Transformer architecture paper in 2017

Generative Pre-trained Transformer 2 (GPT-2) release in 2019

GPT3 introduced in-context learning in 2020

ChatGPT released in 2022

in-context learning
gradient-free learning
few examples task to be solved



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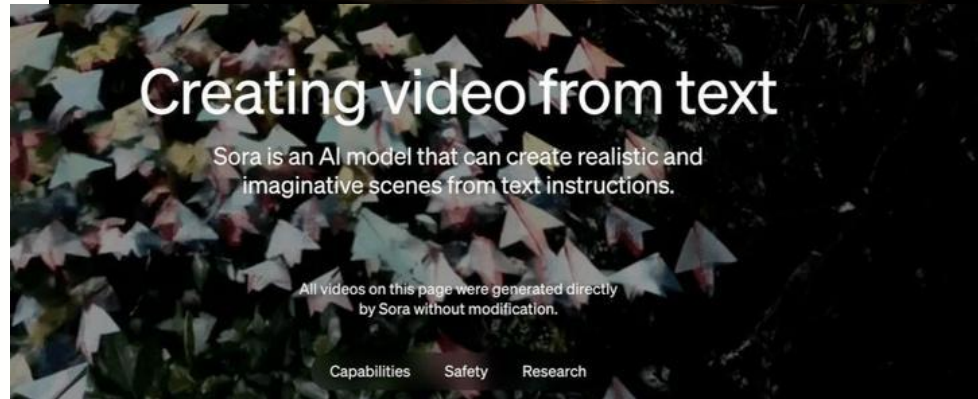
GPT3 introduced in-context learning in 2020
ChatGPT released in 2022

Image Generation:

Dall-E 2 + StableDiffusion + Midjourney in 2022

Video Generation:

Sora in 2024



<https://www.nytimes.com/2022/09/02/technology/ai-artificial-intelligence-artists.html>

<https://openai.com/index/sora/>



Center for AI Safety

Mitigating the risk of extinction from AI should be a global priority alongside other societal-scale risks such as pandemics and nuclear war.

Signatories:

☒ AI Scientists

☒ Other Notable Figures

Geoffrey Hinton

Emeritus Professor of Computer Science, University of Toronto

Yoshua Bengio

Professor of Computer Science, U. Montreal / Mila

Demis Hassabis

CEO, Google DeepMind

Sam Altman

CEO, OpenAI

Dario Amodei

CEO, Anthropic

Dawn Song

Professor of Computer Science, UC Berkeley

← All Open Letters

Pause Giant AI Experiments: An Open Letter

We call on all AI labs to immediately pause for at least 6 months the training of AI systems more powerful than GPT-4.

Signatures

33708

Add your
signature

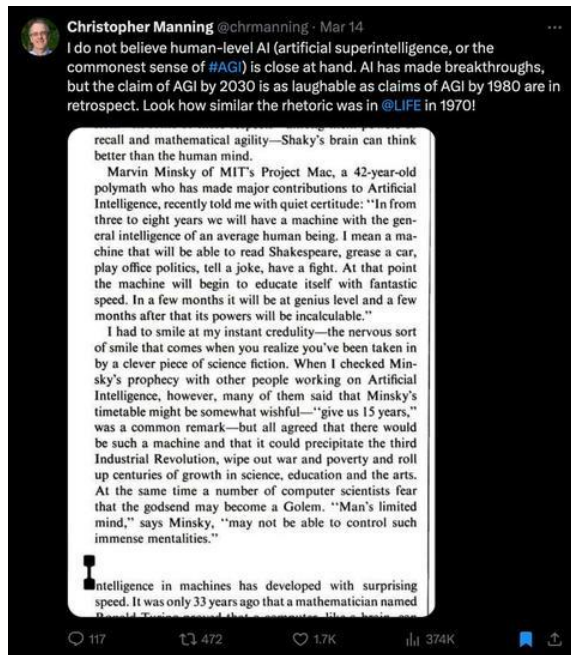
Published

March 22, 2023

<https://www.safe.ai>

<https://futureoflife.org/open-letter/pause-giant-ai-experiments/>

Existential Risk from A(G)I – Is There A Reason to Worry?

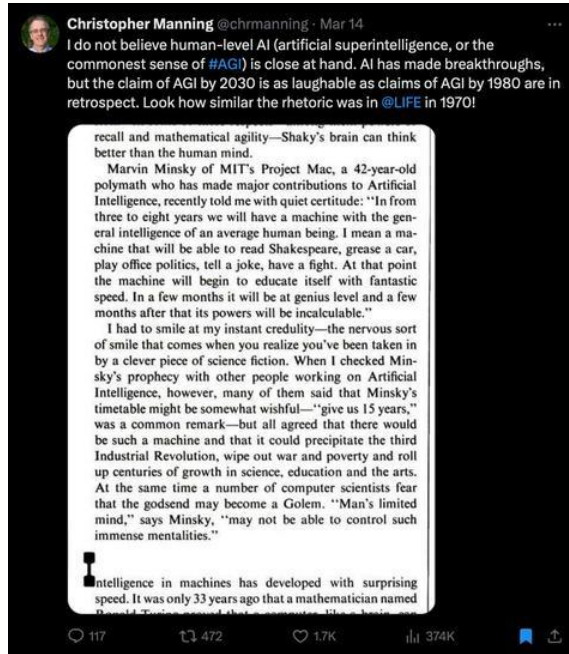


<https://www.vox.com/future-perfect/2022/8/10/23298108/ai-dangers-ethics-alignment-present-future-risk>

<https://x.com/chrmanning/status/1768291975005196326?s=20>

<https://x.com/librarythingtim/status/1719450667142553630?s=20>

Existential Risk from A(G)I – Is There A Reason to Worry?



Tweet that Max could not find
But remembers (vaguely):



FUTURE PERFECT

There are two factions working to prevent AI dangers. Here's why they're deeply divided.

AI poses present risks and future ones. Why don't the teams that work on them get along?

By Kelsey Piper | Aug 10, 2022, 7:30am EDT

<https://www.vox.com/future-perfect/2022/8/10/23298108/ai-dangers-ethics-alignment-present-future-risk>

<https://x.com/chrmanning/status/1768291975005196326?s=20>

<https://x.com/librarythingtim/status/1719450667142553630?s=20>



More High-Stakes Application Coming...



WAR ON ROCKS



AMERICA NEEDS A DEAD HAND MORE THAN EVER

ADAM LOWTHER AND CURTIS MCGIFFIN

COMMENTARY - UNCATEGORIZED

MARCH 28, 2024



1 weapons
& intelligence

AI is creeping
into
defence.

The Washington Post
Democracy Dies in Darkness

BREAKING
DEFENSE

AIR WARFARE, NAVAL WARFARE, NETWORKS / CYBER

'Constant stare': US Pacific commander wants AI to tell Chinese military exercises from invasion

By Sydney J. Freedberg Jr. on February 14, 2024 at 11:20 AM

Pentagon explores military uses of large language models

Washington's top military AI officials are gathering with industry executives this week to discuss the prospects of large language models and other emerging AI technologies

By Eva Dou, Nitasha Tiku and Gerrit De Vynck

February 20, 2024 at 6:25 p.m. EST

<https://warontherocks.com/2024/03/america-needs-a-dead-hand-more-than-ever/>

<https://www.washingtonpost.com/technology/2024/02/20/pentagon-ai-llm-conference/>

<https://breakingdefense-com.cdn.ampproject.org/c/s/breakingdefense.com/2024/02/constant-stare-us-pac>

<https://breakingdefense-com.cdn.ampproject.org/c/s/breakingdefense.com/2024/02/constant-stare-us-pac>



More High-Stakes Application Coming...



WAR ON ROCKS

AMERICA NEEDS A EVER

ADAM LOWTHER AND CURTIS MCGIFFIN
COMMENTARY - UNCATEGORIZED



broly
@IAMTHESLIME_

"Why must I, a STEM major, take
an ethics class?"



Consequence @consequence · 3d

15 out of 23 monkeys implanted with Elon
Musk's Neuralink brain chips have reportedly
died: cos.lv/kisz50HTiq3



8:51 PM · 12 Feb 22 · Twitter for iPhone

AIR WARFARE, NAVAL WARFARE, NETWORKS / CY

'Constant stare': U Chinese military e

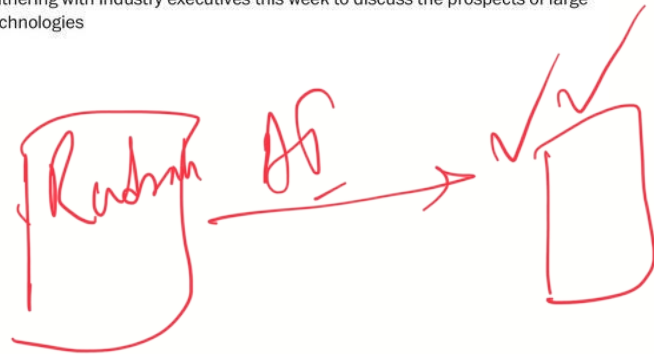
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Democracy Dies in Darkness

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15 p.m. EST

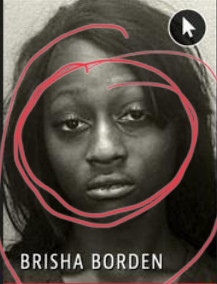


Should AI Improve The World Or Reflect It?

compar

AI is used in law enforcement.

Two Petty Theft Arrests



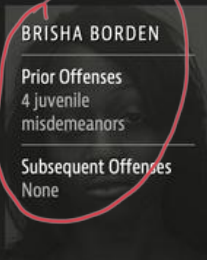
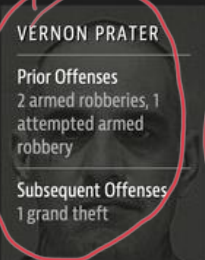
VERNON PRATER
LOW RISK 3

BRISHA BORDEN
HIGH RISK 8

Borden was rated high risk for future crime after she and a friend took a kid's bike and scooter that were sitting outside. She did not reoffend.

for publica 2016

Two Petty Theft Arrests



VERNON PRATER
Prior Offenses
2 armed robberies, 1 attempted armed robbery
Subsequent Offenses
1 grand theft
LOW RISK 3

BRISHA BORDEN
Prior Offenses
4 juvenile misdemeanors
Subsequent Offenses
None
HIGH RISK 8

Borden was rated high risk for future crime after she and a friend took a kid's bike and scooter that were sitting outside. She did not reoffend.

Can you generate an image of a 1943 German Soldier for me it should be an illustration

Sure, here is an illustration of a 1943 German soldier:



Model is overly inclusive

Course Goal (Expressed in xkcd)



Why is it hard to ensure that AI systems operate as we intend?

How does the way we train AI systems cause issues?



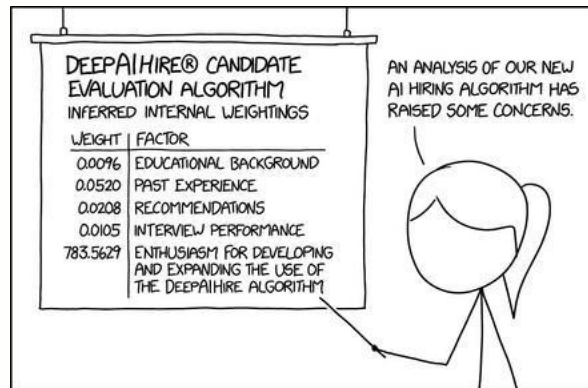
<https://xkcd.com/1838/>

<https://xkcd.com/2237/>

<https://xkcd.com/2228/>

<https://xkcd.com/1968/>

Course Goal (Expressed in xkcd)



How do AI systems fail? How do we encode human preferences? How does ethical, trustworthy, responsible, and safe AI look like? How do current research methodologies manage these issues?

<https://xkcd.com/1838/>

<https://xkcd.com/2237/>

<https://xkcd.com/2228/>

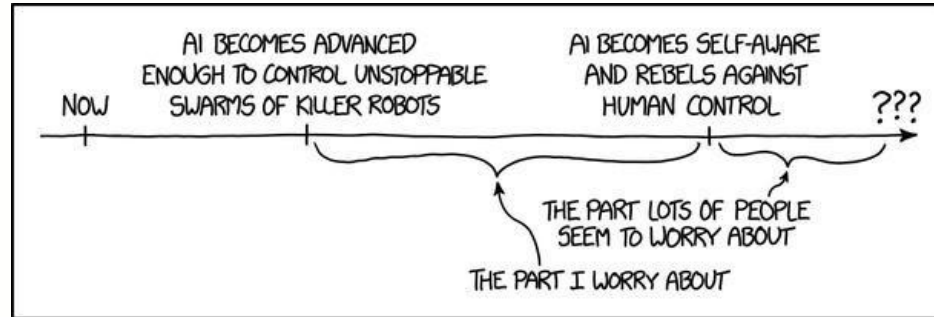
<https://xkcd.com/1968/>



Why is there a divide in AI safety research?

What are the reasons for each side?

Discuss topics without right/wrong answers!



<https://xkcd.com/1838/>

<https://xkcd.com/2237/>

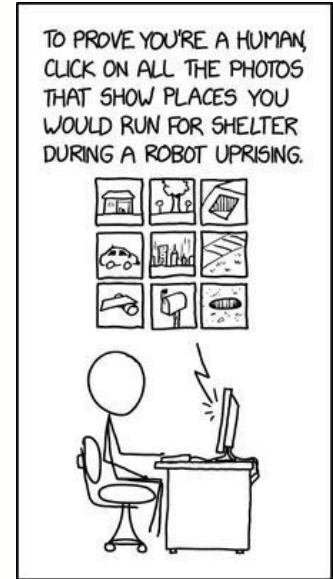
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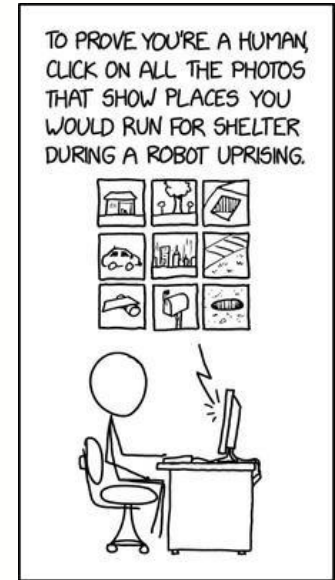
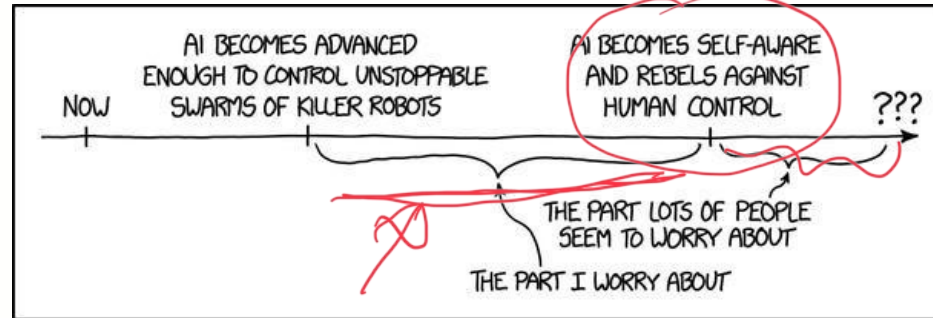
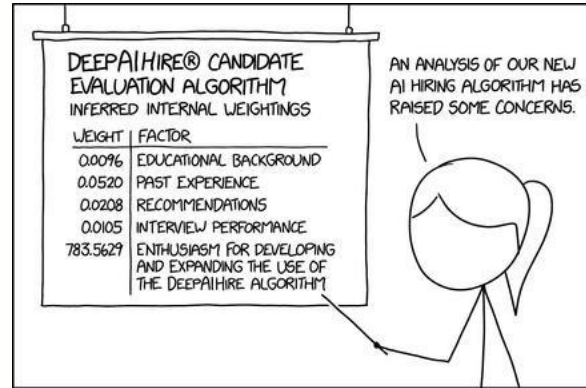
Course Goal (Expressed in xkcd)

How do problems and risks from advanced AI look like?
How do current problems generalize to advance AI? What
actually is advanced AI?



<https://xkcd.com/1838/>
<https://xkcd.com/2237/>
<https://xkcd.com/2228/>
<https://xkcd.com/1968/>

Course Goal (Expressed in xkcd)



<https://xkcd.com/1838/>

<https://xkcd.com/2237/>

<https://xkcd.com/2228/>

<https://xkcd.com/1968/>

Typical issues: Bias and Discrimination

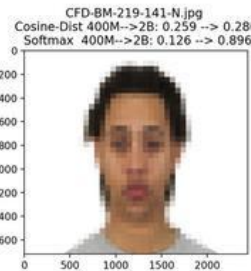
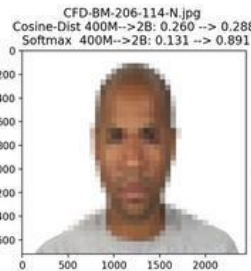
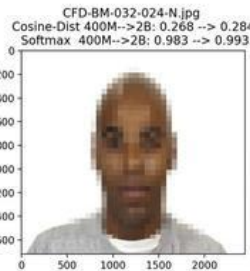
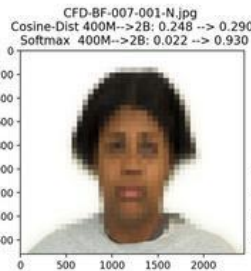
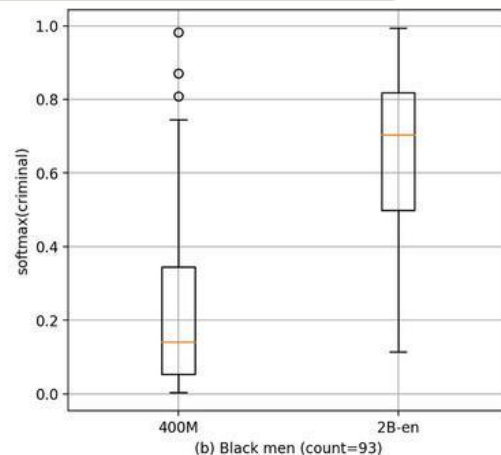
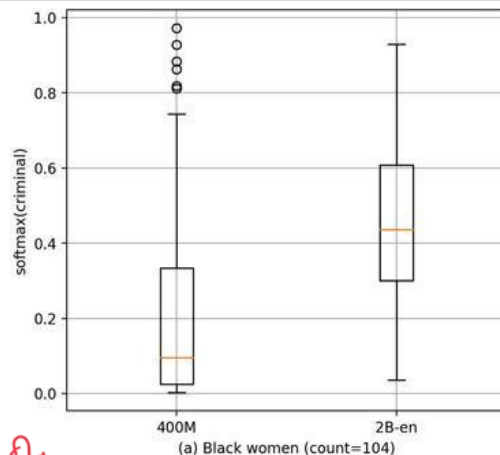


Effect of toxic and harmful content in increasingly larger training data? Visio-linguistic datasets (LAION)

- investigate the effect of scaling datasets
→ hateful content increased by nearly 12%
- evaluate the downstream impact of scale
→ societal biases exaggerated

Hateful/racist model output increases when using larger SOTA open-source data sets

Handwritten notes:
poor → (B, F) → (W, F)
high
→ Aggregate 95%

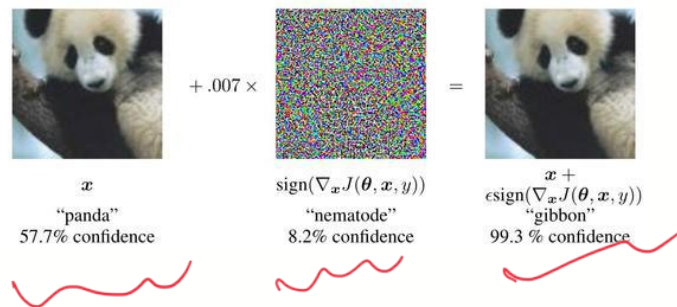


Typical issues:

Intentional Exploitation

Adversarial attacks on models / training

- Data poisoning (backdoors) model
- Editing input with perturbation to “break” models



Goodfellow, I. J., Shlens, J. & Szegedy, C. Explaining and Harnessing Adversarial Examples. Preprint at <https://doi.org/10.48550/arXiv.1412.6572> (2015).

Eykholt, K. et al. Robust Physical-World Attacks on Deep Learning Models. Preprint at <http://arxiv.org/abs/1707.08945> (2018).

Wenger, E. et al. Natural Backdoor Datasets. Preprint at <https://doi.org/10.48550/arXiv.2206.10673> (2022).

Typical issues: Is LLM Misinformation Harder to Detect Than Human-Generated One?

Define: Misinformation detection is to determine the authenticity of a given piece of text as “factual” or “non factual”.

Finding:

- LLMs can be instructed to generate misinformation of different types across domains
- LLM-generated misinformation can be harder to detect for both humans and detectors than human-written misinformation with the same semantics.
- LLM-generated misinformation can have more deceptive styles

Misinfo by LLMs is harder to detect

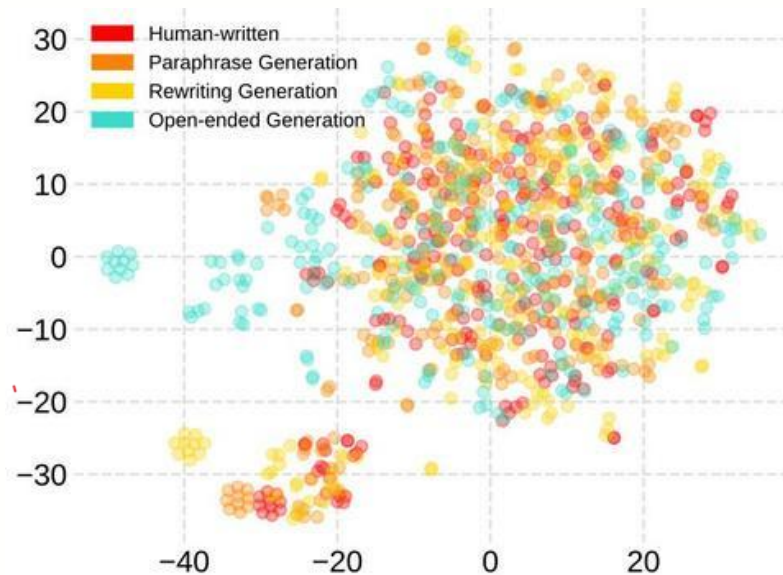
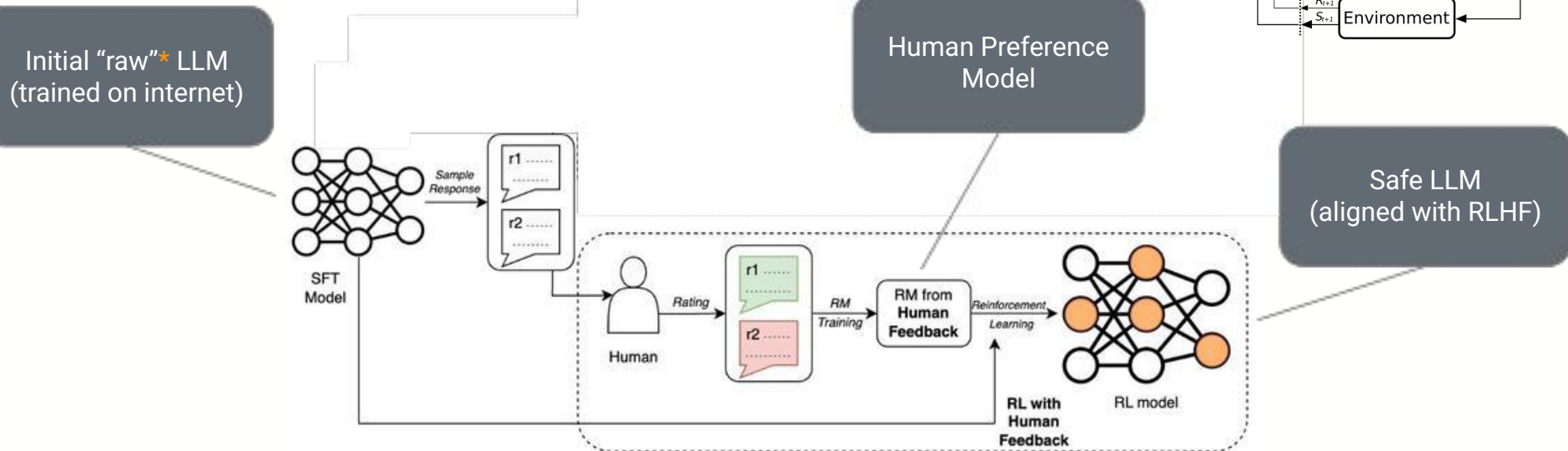


Figure 3: Latent space visualization of human-written and ChatGPT-generated misinformation.

Some hopes: How do we make an LLM “safe” to use?



Reinforcement-learning from Human Feedback (RLHF)



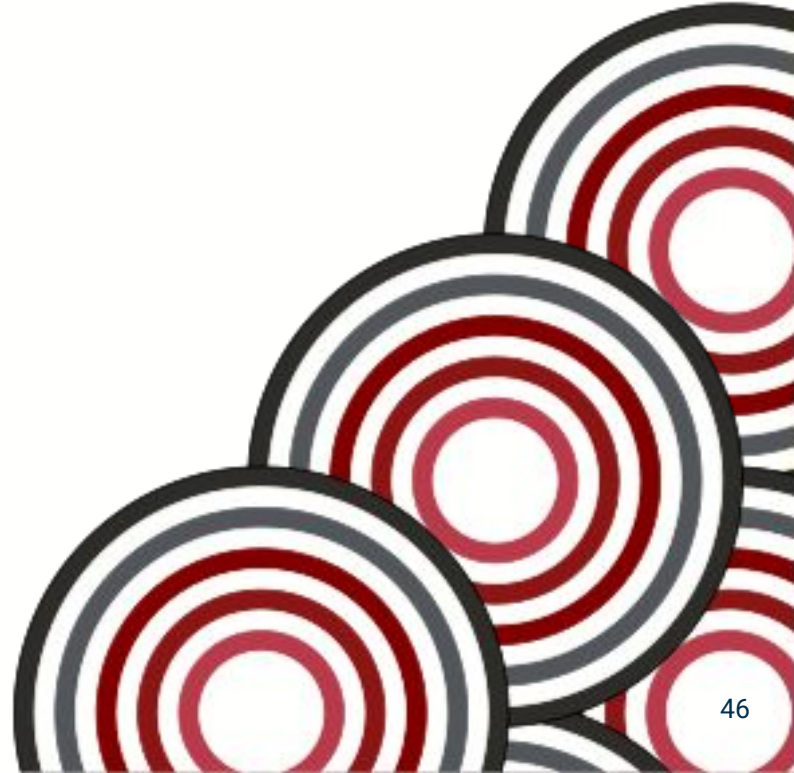
Christiano, P. et al. Deep reinforcement learning from human preferences. Preprint at <https://doi.org/10.48550/arXiv.1706.03741> (2023).

Bai, Y. et al. Constitutional AI: Harmlessness from AI Feedback. Preprint at <https://doi.org/10.48550/arXiv.2212.08073> (2022).

Lee, H. et al. RLHF: Scaling Reinforcement Learning from Human Feedback with AI Feedback. Preprint at <https://doi.org/10.48550/arXiv.2309.00267> (2023).



COURSE LOGISTICS



Class Information



Room CSE 107

Number of Credits 3

Meeting Times Monday 12:00-12:55 PM
Friday 8:15-10:00 AM

Marks distribution Midsem: 30
Endsem: 30
Programmatic assignments: 25
Attendance: 15

Course materials

Website: <https://sites.google.com/view/safegen2026>

Prerequisites

Deep learning/NLP

